

Study Report: Salesforce CRM Developer / Administrator - Career Path & Core Skills

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Family:	Salesforce CRM Developer / Administrator
Level:	Beginner to Advanced
Tracks:	Beginner, Intermediate, Advanced
Chapters:	18
Source:	Official docs, free libraries, and industry-standard references (see sources and books)

Official sources and free libraries

- <https://trailhead.salesforce.com/>
- <https://developer.salesforce.com/docs>
- <https://help.salesforce.com/s/>
- <https://architect.salesforce.com/well-architected>
- <https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/>
- <https://developer.salesforce.com/docs/component-library/documentation/en/lwc>

Summary

Salesforce CRM career path from Trailhead fundamentals through declarative automation (Flows), Apex and triggers, Lightning Web Components, security model, data model design, and integration patterns. Targets Admin, Platform Developer, and Consultant roles using official Salesforce documentation and Well-Architected guidance.

Skill stack by level

Recommended books

1. Salesforce Admins Handbook - Paul M. Kowalski

Level: Beginner | Objects, fields, security, flows, and Lightning App Builder.

2. Trailhead - Admin Beginner Trail - Salesforce

Level: Beginner | Free interactive modules for CRM admin certification path.

3. Advanced Apex Programming - Dan Appleman

Level: Advanced | Bulk-safe Apex, triggers, and governor limit patterns.

4. Salesforce Lightning Web Components - Salesforce Developers

Level: Intermediate | Official LWC guide for custom UI on the Salesforce platform.

5. Salesforce Certified Administrator Study Guide - Salesforce Ben

Level: Intermediate | Exam-focused admin prep: security, automation, and reports.

End-to-end projects

Project 1: Nonprofit Donor Tracker

Beginner | 2-3 weeks - Custom objects for Donors and Donations, page layouts, reports, and a simple Flow for thank-you email.

- Objects: Object schema doc
- UI: Layout screenshots
- Report: Report + dashboard
- Flow: Flow demo video

Project 2: Sales Pipeline Automation

Intermediate | 3-4 weeks - Lead conversion, opportunity stages, approval process for discounts, and Lightning dashboard.

- Leads: Conversion demo
- Pipeline: Stage config doc
- Approval: Approval demo video
- Dashboard: Dashboard screenshot

Project 3: LWC Custom Component Suite

Advanced | 4-5 weeks - Build 3 Lightning Web Components with Apex controllers, Jest tests, and deploy via Salesforce CLI.

- LWC: Components in org
- Apex: Apex test results
- Jest: Jest coverage report
- CI: CI pipeline green

Chapters

Track: Beginner

Chapter 1: Salesforce Platform and Trailhead Navigation [Beginner]

Chapter focus: Salesforce Platform and Trailhead Navigation** *(Level: Beginner)

Salesforce is a multi-tenant cloud CRM platform. Orgs contain standard and custom objects, apps, tabs, and setup menus. Trailhead modules provide free hands-on learning with playgrounds. Understand production vs sandbox vs scratch org lifecycle.

Code Reference:

```
// Open Developer Console: Setup -> Developer Console
// Query: SELECT Id, Name FROM Account LIMIT 5
```

What it shows: SOQL in Developer Console validates data during admin tasks.

Topic guide

What it actually is

Salesforce is the leading cloud CRM and low-code app platform.

When to use it

Sales, service, marketing, and custom business app delivery.

Where to use it

Enterprises, nonprofits, and ed-tech companies running student CRM.

Real use example

CodeQuest learner completes Admin Beginner trail before touching client sandbox.

Chapter 2: Standard Objects and Data Model [Beginner]

Chapter focus: Standard Objects and Data Model** *(Level: Beginner)

Core objects: Lead, Account, Contact, Opportunity, Case, Campaign. Relationships: lookup vs master-detail (rollup implications). Activities (Tasks, Events) polymorphically relate to parents. Design for reporting-excessive lookups hurt performance.

Code Reference:

```
SELECT Id, Name, (SELECT Id, LastName FROM Contacts) FROM Account WHERE Industry = 'Education'
```

What it shows: Subqueries retrieve related contacts for account hierarchy reports.

Topic guide

What it actually is

The data model structures how business entities relate in CRM.

When to use it

Every Salesforce implementation and integration mapping exercise.

Where to use it

B2B sales, customer support, and partner management.

Real use example

Education Account links Contacts (teachers) and Opportunities (district deals).

Chapter 3: Security Model: Profiles and Permission Sets [Beginner]

Chapter focus: Security Model: Profiles and Permission Sets** *(Level: Beginner)

Profiles define baseline object and field permissions; permission sets grant additive access. Use permission set groups for role bundles. Field-Level Security hides sensitive columns; UI does not bypass FLS-Apex must enforce stripInaccessible.

Code Reference:

```
// Check FLS in Apex:  
// SObjectAccessDecision decision = Security.stripInaccessible(AccessType.READABLE,  
records);
```

What it shows: stripInaccessible enforces field-level security in Apex queries.

Topic guide

What it actually is

Salesforce security is layered: org, object, field, record.

When to use it

User onboarding, compliance, and least-privilege reviews.

Where to use it

Every org with more than one user persona.

Real use example

Finance permission set grants invoice fields without cloning admin profile.

Chapter 4: Sharing Rules and Record Access [Beginner]

Chapter focus: Sharing Rules and Record Access** *(Level: Beginner)

OWD (org-wide defaults) set baseline private/public read/write. Sharing rules, roles, teams, and manual sharing widen access. Apex sharing keywords: with sharing, without sharing, inherited sharing. Run sharing recalc after bulk loads.

Code Reference:

```
SELECT Id, Name, AccessLevel FROM AccountShare WHERE AccountId = '001...'
```

What it shows: AccountShare queries debug why users cannot see records.

Topic guide

What it actually is

Record-level sharing controls who sees which rows.

When to use it

Private sales orgs, partner portals, and case segregation.

Where to use it

Multi-region sales teams and support queues.

Real use example

APAC reps see only APAC accounts via role hierarchy and territory rules.

Chapter 5: Reports, Dashboards, and List Views [Beginner]

Chapter focus: Reports, Dashboards, and List Views** *(Level: Beginner)

Reports summarize CRM data; dashboards visualize KPIs for executives. Report types define join paths. Subscribe to dashboard refreshes. List views power daily rep workflows-pin favorites for My Open Opportunities.

Code Reference:

```
SELECT StageName, COUNT(Id) cnt FROM Opportunity GROUP BY StageName
```

What it shows: Aggregate SOQL mirrors report summarization logic.

Topic guide

What it actually is

Reports translate CRM data into decisions without code.

When to use it

Pipeline reviews, service level reviews, and campaign ROI.

Where to use it

Sales managers, support leads, and marketing ops.

Real use example

Dashboard shows CodeQuest pilot conversion rate dropped-triggers campaign review.

Chapter 6: Validation Rules and Page Layouts [Beginner]

Chapter focus: Validation Rules and Page Layouts** *(Level: Beginner)

Validation rules block bad data at save time with formula errors. Page layouts control field visibility per profile. Dynamic forms (UI API) modernize layout per record type. Combine validation with Flow for cross-field checks.

Code Reference:

```
AND(ISBLANK(Phone), ISBLANK(Email))  
// Error: Provide Phone or Email
```

What it shows: Validation formulas enforce contactability on lead capture.

Topic guide

What it actually is

Declarative validation prevents garbage data entering CRM.

When to use it

Data quality initiatives and required field policies.

Where to use it

Lead capture forms, opportunity closure gates.

Real use example

Cannot close Won opportunity without Contract signed checkbox-validation enforces.

Track: Intermediate

Chapter 7: Flow Builder and Automation Strategy [Intermediate]

Chapter focus: Flow Builder and Automation Strategy** *(Level: Intermediate)

Record-triggered Flows replace Workflow Rules and Process Builder. Use before-save flows for field updates without extra DML. Screen flows guide users through multi-step wizards. Document automation inventory-overlapping flows cause recursion errors.

Code Reference:

```
// Flow: Record-Triggered After Save on Opportunity
// Decision: Stage = Closed Won -> Create Task
```

What it shows: Flow decision elements branch automation without Apex.

Topic guide

What it actually is

Flow is the primary declarative automation tool on Salesforce.

When to use it

Assignment, notifications, approvals, and integrations.

Where to use it

Most admin and consultant day-to-day configuration.

Real use example

Closed Won flow creates onboarding task and emails customer success.

Chapter 8: Apex Triggers and Bulkification [Intermediate]

Chapter focus: Apex Triggers and Bulkification** *(Level: Intermediate)

Triggers fire before/after insert/update/delete. Bulkify: never SOQL or DML inside loops over `Trigger.new`. Use handler pattern separating logic from trigger stub. Governor limits: 100 SOQL, 150 DML rows per transaction-design accordingly.

Code Reference:

```
trigger OpportunityTrigger on Opportunity (after update) {
    OpportunityHandler.onAfterUpdate(Trigger.new, Trigger.oldMap);
}
```

What it shows: One-line trigger delegates to testable handler class.

Topic guide

What it actually is

Apex triggers implement complex logic Flow cannot express.

When to use it

Cross-object updates, callouts, and fine-grained control.

Where to use it

Enterprise orgs with custom business rules.

Real use example

Handler aggregates opportunity line changes-one DML update for 200 rows.

Chapter 9: SOQL, SOSL, and Governor Limits [Intermediate]

Chapter focus: SOQL, SOSL, and Governor Limits *(Level: Intermediate)**

SOQL queries structured data; SOSL searches text across objects. Selective queries use indexed fields (Id, Name, foreign keys). Avoid selective filter failures on custom fields without indexes. Query plan tool explains performance.

Code Reference:

```
List<Account> accts = [SELECT Id, Name FROM Account WHERE Industry = 'Education' LIMIT 200];
```

What it shows: LIMIT clause prevents accidental full-table scans in loops.

Topic guide**What it actually is**

Efficient queries keep Apex within governor limits.

When to use it

Every Apex class touching database records.

Where to use it

Triggers, batch jobs, and LWC @wire adapters.

Real use example

Unindexed filter on custom status field times out-admin requests custom index.

Chapter 10: Lightning Web Components Basics [Intermediate]

Chapter focus: Lightning Web Components Basics *(Level: Intermediate)**

LWC uses HTML templates, JavaScript modules, and CSS scoped to components. @api exposes properties; @track reactive state. Use lightning-datatable, navigation mixin, and lightning/uiRecordApi for standard UX consistency.

Code Reference:

```
import { LightningElement, api } from 'lwc';
```

```
export default class CourseCard extends LightningElement {
  @api courseName;
}
```

What it shows: @api properties pass data from parent App Builder pages.

Topic guide

What it actually is

LWC is Salesforce's modern UI framework for custom UX.

When to use it

Custom record actions, dashboards, and Experience Cloud.

Where to use it

Replacing legacy Visualforce and Aura components.

Real use example

CourseCard LWC embeds on Account page showing enrolled CodeQuest modules.

Chapter 11: Integration with REST and Bulk API [Intermediate]

Chapter focus: Integration with REST and Bulk API** *(Level: Intermediate)

REST API handles CRUD on sObjects with OAuth JWT or username-password flow (discouraged). Bulk API 2.0 ingests millions of rows asynchronously. Composite API batches dependent requests. Respect API limits and idempotency for retries.

Code Reference:

```
HttpRequest req = new HttpRequest();
req.setEndpoint('callout:ERP_Named_Credential/accounts');
req.setMethod('GET');
```

What it shows: Named Credentials store auth for secure callouts from Apex.

Topic guide

What it actually is

APIs connect Salesforce to ERP, LMS, and data warehouses.

When to use it

Nightly syncs, real-time notifications, and MDM integrations.

Where to use it

Bi-directional student enrollment between LMS and Salesforce.

Real use example

Bulk API loads 500k nightly rows from data lake into custom staging object.

Chapter 12: Experience Cloud and Portals [Intermediate]

Chapter focus: Experience Cloud and Portals *(Level: Intermediate)**

Experience Cloud sites expose CRM data to partners and customers with branded UX. Configure sharing sets, guest user record access (tighten carefully), and CMS content. LWC runs in Experience Builder pages.

Code Reference:

```
// Guest user: minimal object access + CAPTCHA on public forms
```

What it shows: Security comments remind architects of guest user risks.

Topic guide

What it actually is

Experience Cloud delivers external-facing CRM experiences.

When to use it

Partner deal registration, customer support, and student portals.

Where to use it

B2B ecosystems and education community sites.

Real use example

District admins view support cases in portal without full internal license.

Chapter 13: Deployment with Salesforce DX [Intermediate]

Chapter focus: Deployment with Salesforce DX *(Level: Intermediate)**

SFDX uses source control, scratch orgs, and manifest package.xml. sf project deploy start pushes metadata; CI runs Apex tests on pull requests. Never change prod directly-pipeline promotes dev ? UAT ? prod with change sets or pipelines.

Code Reference:

```
sf org login web --alias uat  
sf project deploy start --target-org uat --test-level RunLocalTests
```

What it shows: CLI deploy with tests mirrors enterprise ALM gates.

Topic guide

What it actually is

Salesforce DX modernizes metadata lifecycle management.

When to use it

Team development and regulated release processes.

Where to use it

ISVs and internal centers of excellence.

Real use example

Pull request fails on 78% coverage-developer adds handler tests before merge.

Track: Advanced

Chapter 14: Asynchronous Apex Patterns [Advanced]

Chapter focus: Asynchronous Apex Patterns** *(Level: Advanced)

Queueable, Batch Apex, and Schedulable process large volumes beyond synchronous limits. Batch implements Database.Batchable with scope chunks. Platform Events decouple publishers and subscribers for event-driven scale.

Code Reference:

```
public class EnrollmentBatch implements Database.Batchable<SObject> {
    public Database.QueryLocator start(Database.BatchableContext bc) {
        return Database.getQueryLocator('SELECT Id FROM Enrollment__c');
    }
}
```

What it shows: Batch skeleton processes custom enrollment objects nightly.

Topic guide

What it actually is

Async Apex handles enterprise data volume outside user transactions.

When to use it

Nightly syncs, rollups, and data archival.

Where to use it

Orgs exceeding synchronous governor thresholds.

Real use example

Batch recalculates district utilization scores for 2M child records.

Chapter 15: Enterprise Integration Architecture [Advanced]

Chapter focus: Enterprise Integration Architecture** *(Level: Advanced)

Apply Salesforce Well-Architected: reliable, secure, adaptable. Use anti-corruption layers, outbox patterns, and idempotent consumers. MuleSoft or Event Relay for hybrid buses.

Code Reference:

```
// Change Data Capture subscriber processes Account updates to ERP
```

What it shows: CDC comment illustrates event-driven integration style.

Topic guide

What it actually is

Integration architecture prevents brittle point-to-point spaghetti.

When to use it

Multi-system landscapes with ERP, LMS, and billing.

Where to use it

Large enterprises and acquisition-heavy portfolios.

Real use example

CDC streams opportunity wins to provisioning microservice within seconds.

Chapter 16: Advanced Security and Shield [Advanced]

Chapter focus: Advanced Security and Shield *(Level: Advanced)**

Shield Platform Encryption, Event Monitoring, and Field Audit Trail address compliance. Transaction Security policies block CSV exports by profile. Monitor Login Forensics for credential stuffing. Tenant secret rotation requires planning.

Code Reference:

```
// Event Monitoring: ReportEvent for bulk data export detection
```

What it shows: ReportEvent tracks who exported large contact lists.

Topic guide**What it actually is**

Advanced security features protect regulated CRM data.

When to use it

Healthcare, financial services, and public sector CRM.

Where to use it

Orgs under HIPAA, PCI, or FedRAMP requirements.

Real use example

Transaction Security blocks guest user SOQL injection attempt.

Chapter 17: Performance Optimization and Caching [Advanced]

Chapter focus: Performance Optimization and Caching *(Level: Advanced)**

Optimize LWC with lazy load and wire cacheable Apex. Server-side: selective SOQL, custom indexes, skinny triggers. Use debug logs and Apex Profiler. Platform Cache stores session and org cache partitions for hot data.

Code Reference:

```
@AuraEnabled(cacheable=true)
public static List<Metric> getMetrics(Id accountId) { ... }
```

What it shows: Cacheable Apex reduces redundant database hits for dashboards.

Topic guide

What it actually is

Performance tuning keeps UX snappy at enterprise user counts.

When to use it

Slow pages, timeout errors, and large-data Lightning apps.

Where to use it

Global sales teams with heavy custom UIs.

Real use example

Cacheable metrics API drops LWC load from 4s to 400ms.

Chapter 18: Einstein and AI on Salesforce [Advanced]

Chapter focus: Einstein and AI on Salesforce** *(Level: Advanced)

Einstein Lead Scoring, Case Classification, and Prompt Builder bring AI into CRM workflows. Understand model bias, human review gates, and data grounding. Agentforce orchestrates actions with audit trails for compliance.

Code Reference:

```
// Einstein Case Classification suggests Reason field on new cases
```

What it shows: Classification suggestions speed agent triage with human override.

Topic guide

What it actually is

Salesforce AI augments reps and agents with predictive and generative features.

When to use it

High-volume service queues and inside sales teams.

Where to use it

Orgs adopting CRM-embedded AI instead of bolt-on tools.

Real use example

Einstein summarizes long Case email threads before CodeQuest agent responds.